Hades Model Report

v0.8.4 GitHub

Hastie & Smith, et al. (2026)

MODEL CONFIGURATION

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

Mode	Alkaline
Submode	-
Data suite	./sao_tome.xlsx
$\mathrm{FeO}/\mathrm{Fe}^{2+}$	0.9
Stages Run	Primitive Basic

1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 65.55 wt.%

	Primitive	Basic	Total
Olivine	28.07	12.68	20.9
Plagioclase	0.31	24.4	11.54
Magnetite	1.32	6.47	3.72
Ilmenite	5.01	3.87	4.48
Titanaugite	64.5	44.71	55.27
Alkali-amphibole	0.8	2.86	1.76
Apatite		5.0	2.33
FC%	35.0	47.0	

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

2. Parameter Configuration

The parameters below define the model as it was executed for this run.

	Primary	Primitive	Basic	Intermediate	Evolved
Plagioclase	1.2	1.2	1.2	1.2	1.2
Clinopyroxene	0.28	0.28	0.28	0.28	0.28
Orthopyroxene	0.29	0.29	0.29	0.29	0.29
Amphibole	0.37	0.37	0.37	0.37	0.37
K-feldspar	1.3	1.3	1.3	1.3	1.3
Biotite	1.0	1.0	1.0	1.0	1.0
Titanaugite	1.0	1.0	1.0	1.0	1.0
Nepheline	1.0	1.0	1.0	1.0	1.0
Leucite	1.0	1.0	1.0	1.0	1.0
Nosean/Hauyne	1.0	1.0	1.0	1.0	1.0
Alkali-amphibole	1.0	1.0	1.0	1.0	1.0

Table 2: Partition coefficients for the model phases.

3. Magma Evolution

Initial Compositions

	Primary	Primitive	Basic	Intermediate	Evolved	End
Sample		ST18	ST96	ST55		
SiO_2		45.38	51.32	59.19		
TiO_2		3.45	2.59	0.36		
Al_2O_3		14.0	17.88	21.04		
Fe_2O_3		12.8	9.87	2.54		
MgO		8.7	4.23	0.28		
CaO		10.16	7.38	1.53		
Na_2O		3.34	3.86	8.98		
K_2O		2.17	2.88	6.07		
P		3326.89	4274.31	183.37		
Ba		766.4	853.9	1558.3		
Rb		57.6	61.6	193.3		
Sr		971.5	1056.4	464.6		
Nb		107.85	119.25	108.19		
Zr		394.5	526.6	751.9		
Y		28.9	36.9	13.6		
La		77.8	89.0	73.8		
Ce		144.0	180.2	94.5		
Nd		61.3	75.8	19.8		
Sm		0.0	0.0	0.0		

	Primary	Primitive	Basic	Intermediate	Evolved	End
Eu		0.0	0.0	0.0		
Gd		0.0	0.0	0.0		
Tb		0.0	0.0	0.0		
Dy		0.0	0.0	0.0		
Ho		0.0	0.0	0.0		
Er		0.0	0.0	0.0		
Tm		0.0	0.0	0.0		
Yb		0.0	0.0	0.0		
Lu		0.0	0.0	0.0		
Hf		0.0	0.0	0.0		
Ta		0.0	0.0	0.0		
Th		7.8	11.9	24.6		
U		0.0	0.0	0.0		

Table 3: Major oxide compositions of the model phases (wt.% for oxides, ppm for trace elements).

Computed Compositions

	Primary	Primitive	Basic	Intermediate	Evolved	End
SiO2		45.38	49.71	61.68		
TiO2		3.45	2.58	0.59		
Al2O3		14.0	17.9	21.83		
Fe2O3		12.8	9.88	3.35		
MgO		8.7	4.22	0.64		
CaO		10.16	7.39	-0.28		
Na2O		3.34	4.98	8.03		
K2O		2.17	3.33	6.16		
P		3326.89	5118.3	945.76		
Ba		766.4	1177.06	2121.78		
Rb		57.6	88.32	162.99		
Sr		971.5	1452.06	957.56		
Nb		107.85	163.64	293.68		
Zr		394.5	585.49	1054.91		
Y		28.9	36.51	25.45		
La		77.8	109.82	65.38		
Ce		144.0	198.76	67.15		
Nd		61.3	78.17	18.32		
Sm		0.0				
Eu		0.0				
Gd		0.0				
Tb		0.0				
Dy		0.0				
Ho		0.0				
Er		0.0				
Tm		0.0				
Yb		0.0				
Lu		0.0				

	Primary	Primitive	Basic	Intermediate	Evolved	End
Hf		0.0				
Ta		0.0				
Th		7.8	11.95	21.41		
U		0.0				

Table 4: Major oxide compositions of the model phases (wt.%).

4. Optimisation Results

Primitive > Basic

PRIMITIVE

Population size	50
Iterations	500/500
Cost function	1.08
Trace analysis?	True

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	45.38	51.32	49.71	3.14
TiO2	3.45	2.59	2.58	0.25
Al2O3	14.0	17.88	17.9	0.1
Fe2O3	12.8	9.87	9.88	0.1
MgO	8.7	4.23	4.22	0.07
CaO	10.16	7.38	7.39	0.22
Na2O	3.34	3.86	4.98	29.14
K2O	2.17	2.88	3.33	15.67
P	3326.89	4274.31	5118.3	19.75
Ba	766.4	853.9	1177.06	37.85
Rb	57.6	61.6	88.32	43.38
Sr	971.5	1056.4	1452.06	37.45
Nb	107.85	119.25	163.64	37.23
Zr	394.5	526.6	585.49	11.18
Y	28.9	36.9	36.51	1.06
La	77.8	89.0	109.82	23.39
Ce	144.0	180.2	198.76	10.3
Nd	61.3	75.8	78.17	3.12
Sm	0.0	0.0		
Eu	0.0	0.0		
Gd	0.0	0.0		
Tb	0.0	0.0		
Dy	0.0	0.0		
Ho	0.0	0.0		

	Initial	Expected	Computed	% Error
Er	0.0	0.0		
Tm	0.0	0.0		
Yb	0.0	0.0		
Lu	0.0	0.0		
Hf	0.0	0.0		
Ta	0.0	0.0		
Th	7.8	11.9	11.95	0.41
U	0.0	0.0		

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	35.0
Olivine	28.07
Spinel	0.0
Plagioclase	0.31
K-feldspar	0.0
Biotite	0.0
Magnetite	1.32
Ilmenite	5.01
Titanaugite	64.5
Nepheline	0.0
Leucite	0.0
Nosean/Hauyne	0.0
Alkali-amphibole	0.8
Apatite	0.0
Zircon	0.0
Monazite	0.0
Allanite	0.0

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	37.26	5.07	6.78	18.53	16.73	15.32	0.28	0.02

Table 7: Bulk mineralogy of the model phases (wt.%).

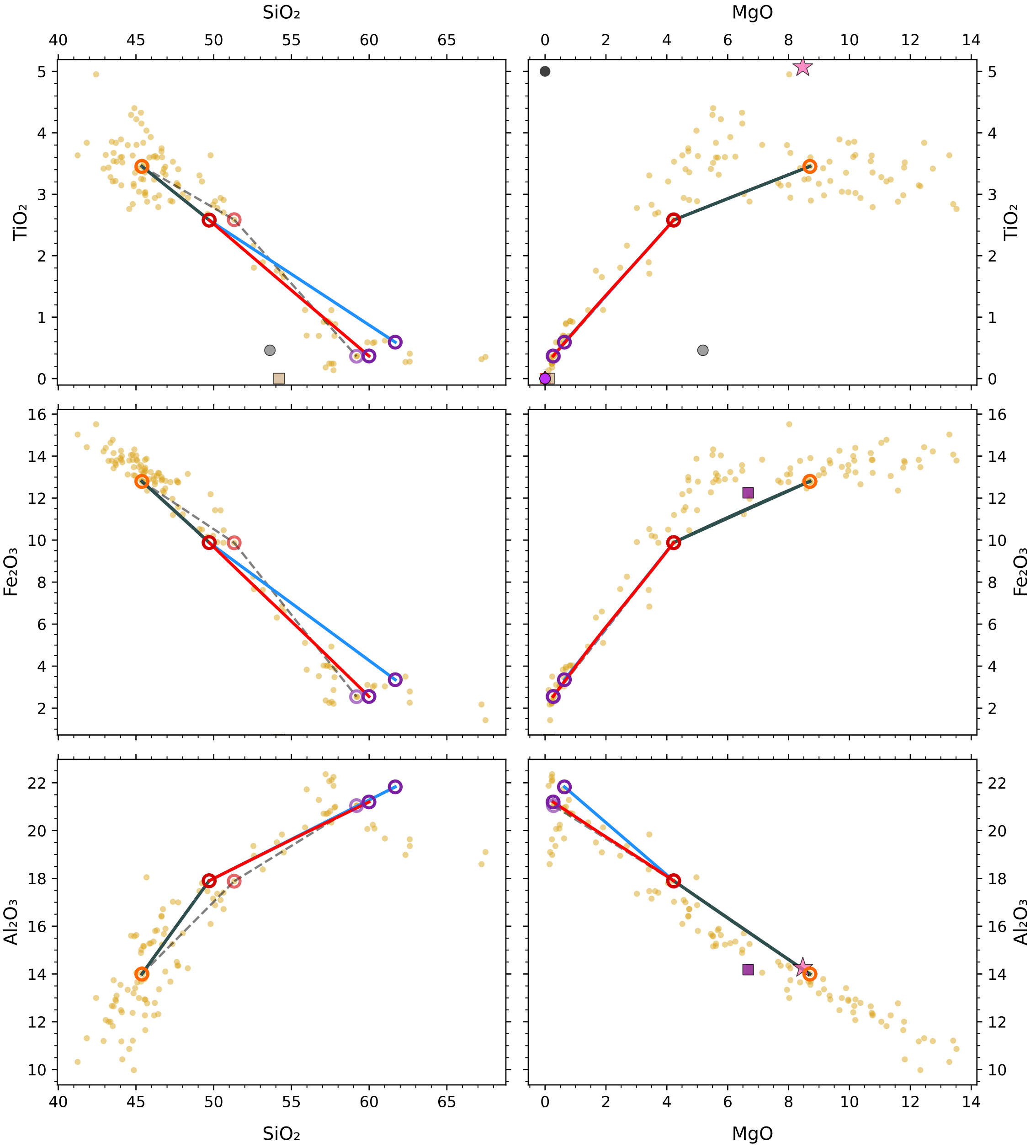
Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Olivine	38.39	0.08	0.19	19.32	41.62	0.4	0.0	0.0
Spinel								
Plagioclase	51.2	0.0	31.09	0.28	0.01	13.98	3.31	0.13
K-feldspar								
Biotite								
Magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
Ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Titanaugite	40.16	3.84	10.27	14.11	7.76	23.5	0.36	0.0
Nepheline								
Leucite								
Nosean/Hauyne								
Alkali-amphibole	53.62	0.46	0.62	31.55	5.19	1.36	5.05	2.15
Apatite								
Zircon								
Monazite								
Allanite								

Table 8: Cotectic compositions of the model phases (wt.% for oxides, ppm for trace elements). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

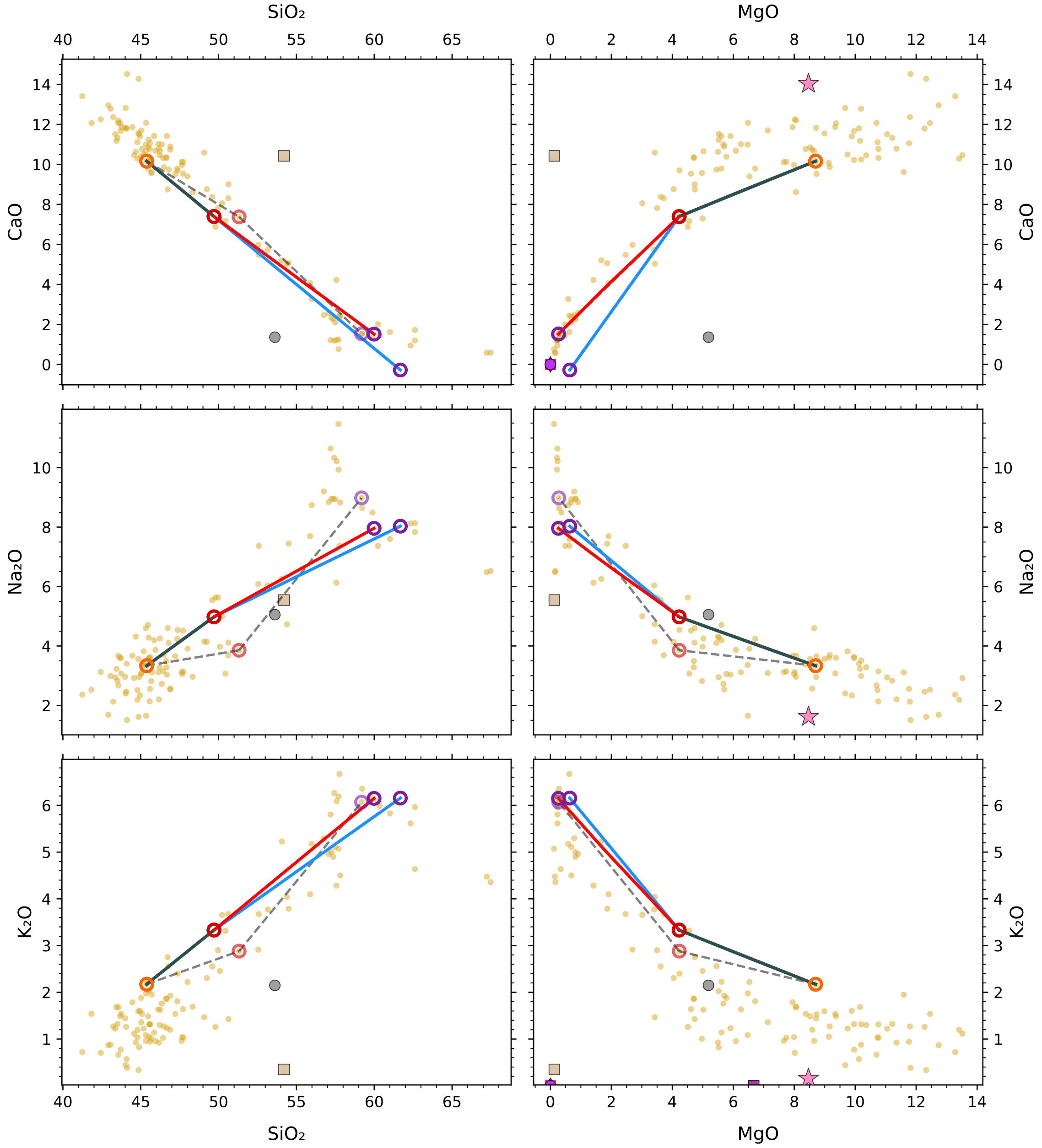


Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primitive to Basic transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Basic > Intermediate

BASIC

Population size	50
Iterations	500/500
Cost function	0.56
Trace analysis?	True

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	49.71	59.19	61.68	4.2
TiO2	2.58	0.36	0.59	62.95

	Initial	Expected	Computed	% Error
Al2O3	17.9	21.04	21.83	3.73
Fe2O3	9.88	2.54	3.35	31.77
MgO	4.22	0.28	0.64	130.07
CaO	7.39	1.53	−0.28	118.26
Na2O	4.98	8.98	8.03	10.61
K2O	3.33	6.07	6.16	1.43
P	5118.3	183.37	945.76	415.76
Ba	1177.06	1558.3	2121.78	36.16
Rb	88.32	193.3	162.99	15.68
Sr	1452.06	464.6	957.56	106.1
Nb	163.64	108.19	293.68	171.45
Zr	585.49	751.9	1054.91	40.3
Y	36.51	13.6	25.45	87.16
La	109.82	73.8	65.38	11.4
Ce	198.76	94.5	67.15	28.94
Nd	78.17	19.8	18.32	7.47
Sm		0.0		
Eu		0.0		
Gd		0.0		
Tb		0.0		
Dy		0.0		
Ho		0.0		
Er		0.0		
Tm		0.0		
Yb		0.0		
Lu		0.0		
Hf		0.0		
Ta		0.0		
Th	11.95	24.6	21.41	12.95
U		0.0		

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	47.0
Olivine	12.68
Spinel	0.0
Plagioclase	24.4
K-feldspar	0.0
Biotite	0.0
Magnetite	6.47
Ilmenite	3.87
Titanaugite	44.71
Nepheline	0.0
Leucite	0.0

	Value
Nosean/Hauyne	0.0
Alkali-amphibole	2.86
Apatite	5.0
Zircon	0.0
Monazite	0.0
Allanite	0.0

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
wt.%	36.09	4.82	13.57	17.57	8.01	16.02	1.54	0.15	9823.5

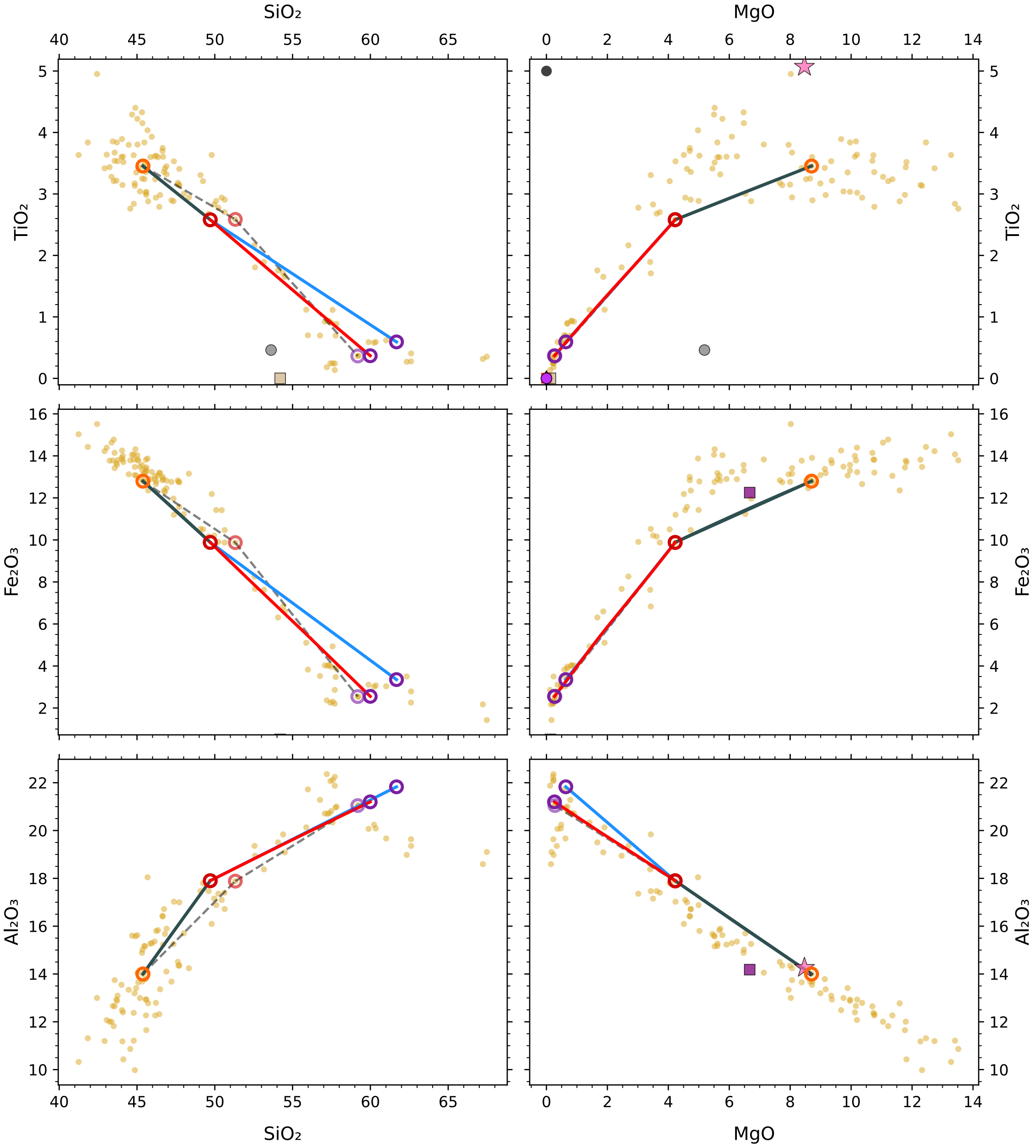
Table 11: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
Olivine	36.92	0.03	1.33	23.48	38.2	0.04	0.0	0.0	0.0
Spinel									
Plagioclase	54.2	0.0	28.84	0.51	0.13	10.42	5.55	0.35	0.0
K-feldspar									
Biotite									
Magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0	0.0
Ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0	0.0
Titanaugite	37.24	5.68	14.18	12.25	6.67	23.88	0.09	0.0	0.0
Nepheline									
Leucite									
Nosean/Hauyne									
Alkali-amphibole	53.62	0.46	0.62	31.55	5.19	1.36	5.05	2.15	0.0
Apatite	0.0	0.0	0.0	0.0	0.0	55.0	0.0	0.0	196470.0
Zircon									
Monazite									
Allanite									

Table 12: Cotectic compositions of the model phases (wt.% for oxides, ppm for trace elements). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



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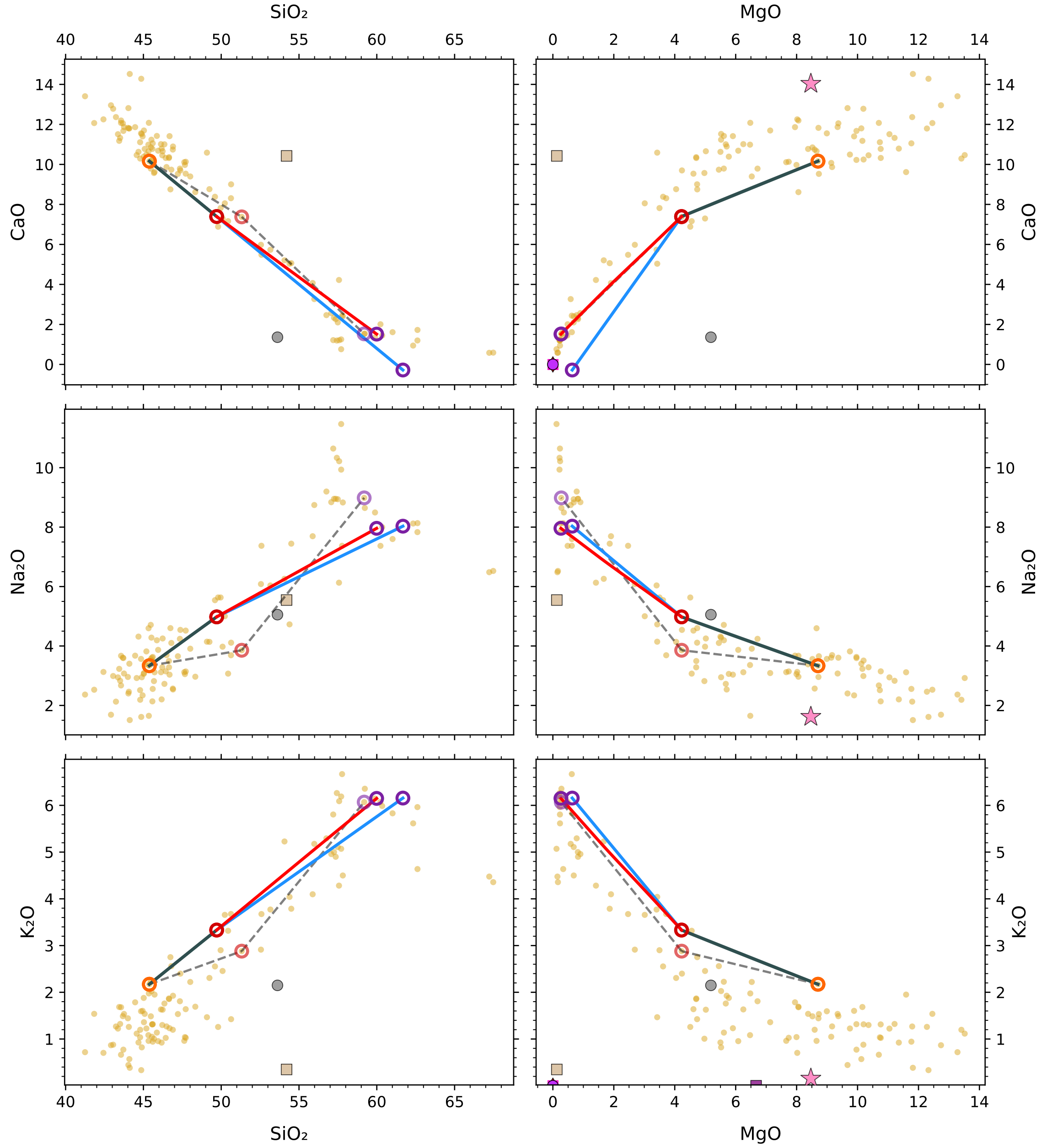

Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Basic to Intermediate transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

5. Figures

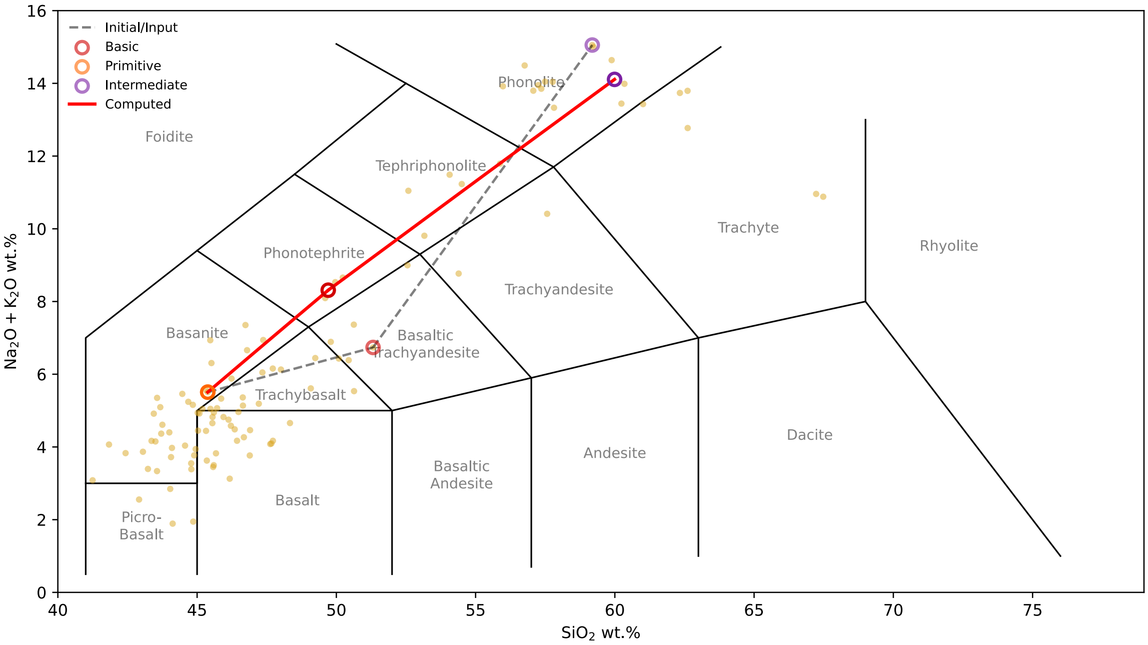

Figure 3: Total Alkali-Silica (TAS) diagram of the model phases.

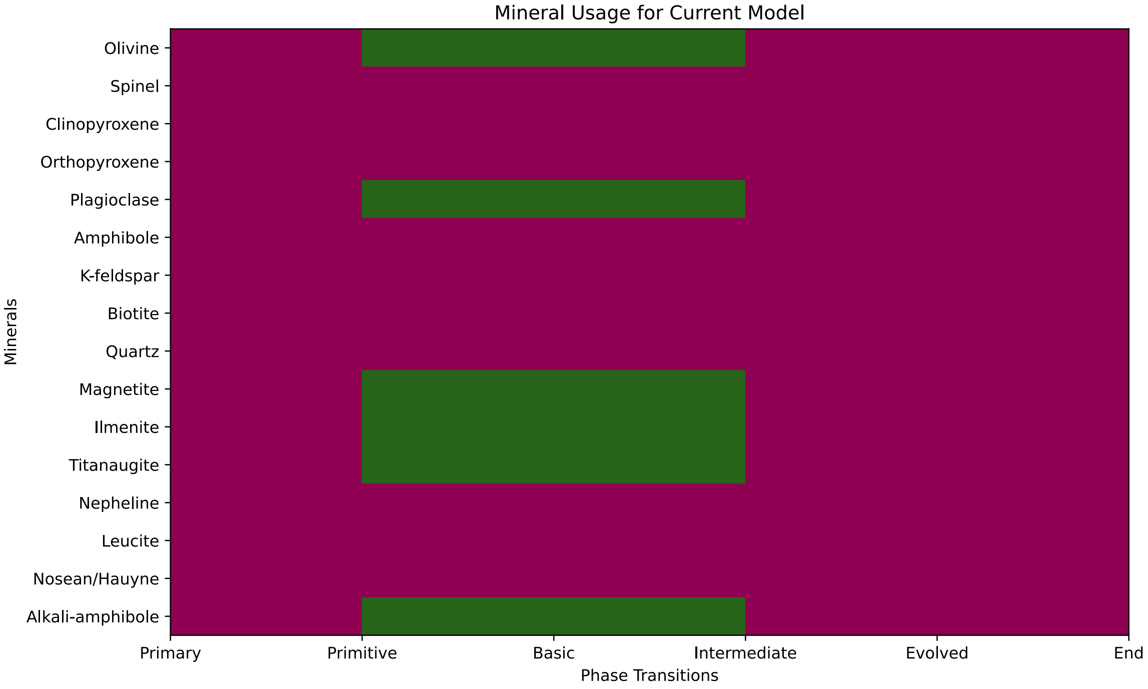

Figure 4: Mineral usage across the model phases.

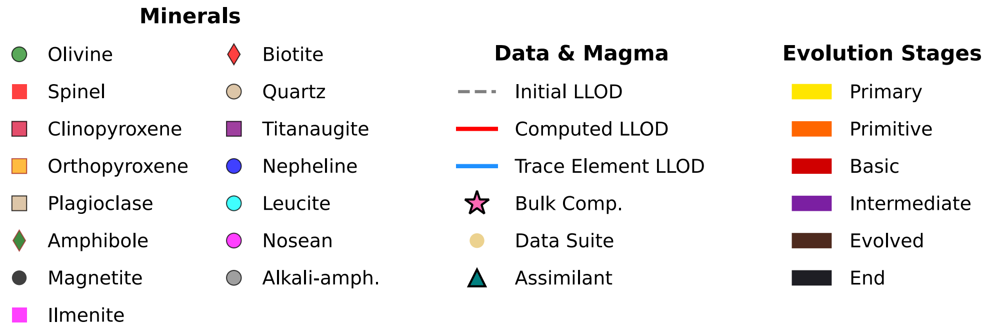
Minerals
Olivine
Spinel
Clinopyroxene
Orthopyroxene
Plagioclase
Amphibole
Magnetite
Ilmenite
Biotite
Quartz
Titanaugite
Nepheline
Leucite
Nosean
Alkali-amph.
Data & Magma
Initial LLOD
Computed LLOD
Trace Element LLOD
Bulk Comp.
Data Suite
Assimilant
Evolution Stages
Primary
Primitive
Basic
Intermediate
Evolved
End

Figure 5: Legend for mineral symbols used in the figures.

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